M2 · Cellular components of innate immunity

Innate immunity lecture pack · EAACI/UEMS Knowledge Exam preparation

After this module you can

- Place each innate cell on the myeloid or lymphoid branch of haematopoiesis and name its key growth factor.
- Compare neutrophils and macrophages, and describe classical (M1) and alternative (M2) macrophage activation.
- Distinguish conventional and plasmacytoid dendritic cells and explain how dendritic cells link innate and adaptive immunity.
- Explain "missing self" and the balance of activating and inhibitory receptors on natural killer (NK) cells.
- Compare mast cells, basophils and eosinophils (receptors, granules, mediators, disease relevance in allergy).
- Match ILC1, ILC2 and ILC3 to their transcription factors, cytokines and functions, and describe the role of ILC2s in allergy.

ADCC	antibody-dependent cell-mediated cytotoxicity
CGD	chronic granulomatous disease
cDC / pDC	conventional / plasmacytoid dendritic cell
CTL	cytotoxic T lymphocyte
DC	dendritic cell
ECP	eosinophil cationic protein
FcεRI	high-affinity Fc receptor for IgE (Fcε receptor type I)
FLT3L	FLT3 ligand
G-CSF / GM-CSF / M-CSF	granulocyte / granulocyte-macrophage / monocyte colony-stimulating factor
HSC	haematopoietic stem cell
IFN	interferon
ILC	innate lymphoid cell
ITAM / ITIM	immunoreceptor tyrosine-based activation / inhibition motif
KIR	killer cell immunoglobulin-like receptor
LAD	leukocyte adhesion deficiency
MAIT	mucosa-associated invariant T cell
MBP	major basic protein
MHC-I	major histocompatibility complex class I
MPO	myeloperoxidase
NCR	natural cytotoxicity receptor
NET	neutrophil extracellular trap
NK	natural killer (cell)
iNKT	invariant natural killer T cell
PAF	platelet-activating factor
PAMP	pathogen-associated molecular pattern
PGD2 / LTC4	prostaglandin D2 / leukotriene C4

PMN	polymorphonuclear leukocyte
SCF	stem cell factor (c-KIT ligand)
SM	systemic mastocytosis
TLR	Toll-like receptor
TSLP	thymic stromal lymphopoietin

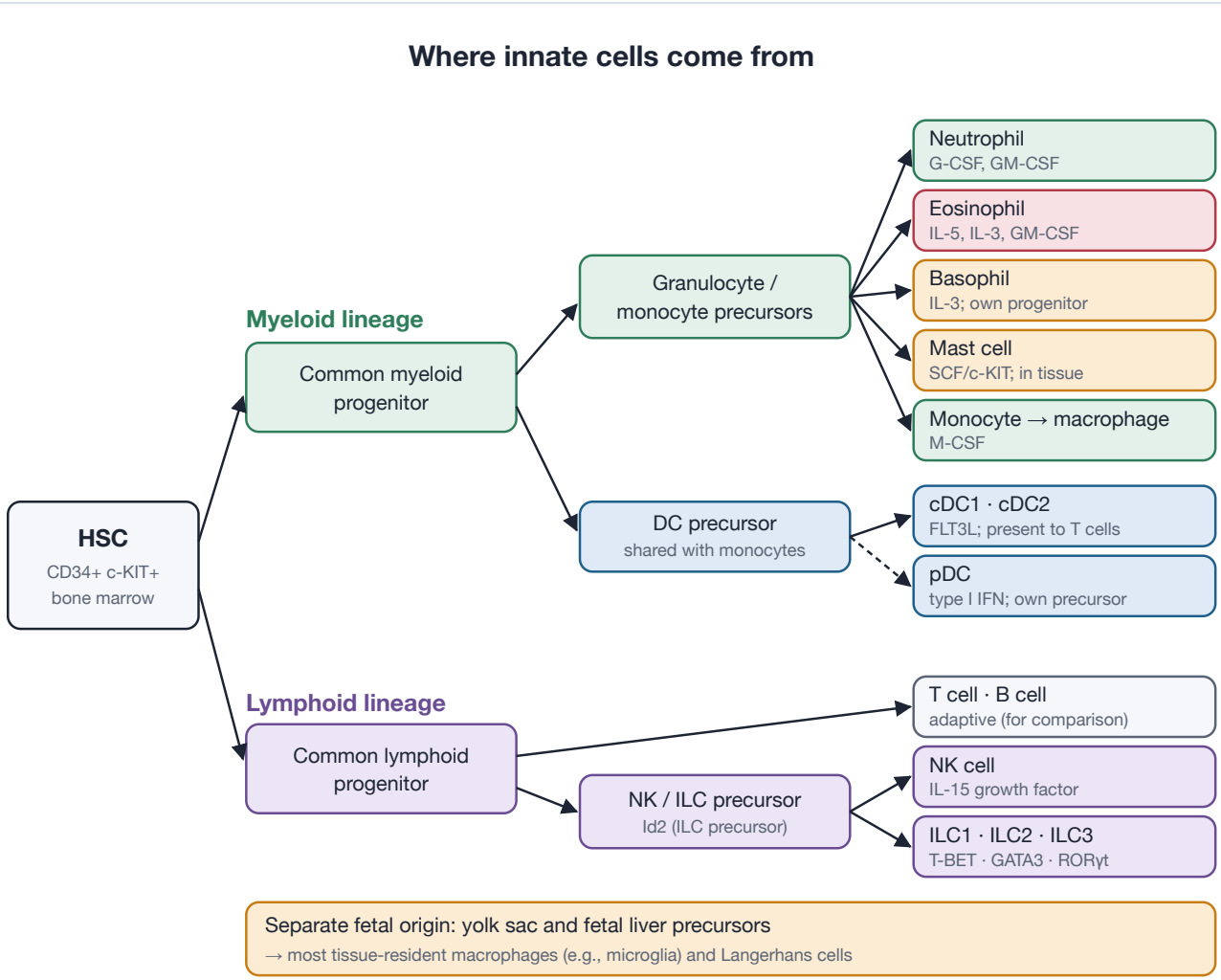
1. Where innate cells come from

All blood cells come from one haematopoietic stem cell (HSC) in the bone marrow. This includes granulocytes, monocytes, dendritic cells (DCs), mast cells, lymphocytes and innate lymphoid cells (ILCs). Abbas p.30 HSCs express CD34 and c-KIT and do not express lineage markers. Abbas p.30

Two branches follow:

- **Myeloid branch.** Common myeloid-megakaryocyte-erythroid progenitors make the granulocytes (neutrophils, eosinophils, basophils) and monocytes. Abbas p.32 Most DCs come from a precursor that also makes monocytes. Abbas p.32 Mast cell progenitors come from a granulocyte/monocyte precursor, leave the marrow, and mature in the tissues. Abbas p.32 Basophils mature in the marrow from progenitors that are different from mast cell progenitors. Abbas p.20
- **Lymphoid branch.** The common lymphoid progenitor makes T cells, B cells and a shared precursor of natural killer (NK) cells and ILCs. Abbas p.30

Key growth factors: G-CSF and GM-CSF drive neutrophil production Abbas p.14; M-CSF drives the monocyte-macrophage lineage Abbas p.15; all DC subsets need FLT3 ligand (FLT3L) and express CD11c. Abbas p.20 In normal blood, neutrophils are 40–60%, eosinophils 1–4%, basophils <1% and monocytes 2–8% of leukocytes. Abbas p.14



D4. Lineage tree of innate cells. Myeloid cells (green, amber, red, blue) and lymphoid cells (purple) come from one HSC. Mast cells leave the marrow as progenitors and mature in tissues. Most tissue-resident macrophages and Langerhans cells come from fetal precursors, not from adult monocytes. Dashed arrow: pDCs develop mainly from their own precursor, distinct from the cDC precursor; only some may arise from the common DC precursor. Fact IDs M2-F001, M2-F002, M2-F003, M2-F004, M2-F005, M2-F010, M2-F006, M2-F007, M2-F008, M2-F009, M2-F082, M2-F066, M2-F048, M2-F093, M2-F094, M2-F022, M2-F039, M2-F114. Abbas p.30 Abbas p.32 Abbas p.20 Abbas p.466 Abbas p.81

Source difference

Abbas (11th ed.) states that most DCs come from a myeloid precursor shared with monocytes. Abbas p.32 Janeway (8th ed.) states that DCs come from both myeloid and lymphoid progenitors. Janeway p.77 Janeway also describes tissue macrophages as cells that mature continuously from monocytes. Janeway p.76 Abbas (newer) states that most tissue-resident macrophages come from yolk sac or fetal liver precursors. Abbas p.15 Use the Abbas version for the exam.

2. Neutrophils

Neutrophils are the most abundant circulating white blood cells and the principal cell of acute inflammation. Abbas p.14 The nucleus has three to five lobes. For this reason, neutrophils are also called polymorphonuclear leukocytes (PMNs). Abbas p.14

Neutrophils have two types of granules:

- **Specific granules** (the majority): lysozyme, collagenase and elastase. Abbas p.14
- **Azurophilic granules**: myeloperoxidase (MPO), defensins and cathelicidins. Abbas p.14

Human neutrophil markers include CD11b, CD18, CD15, CD66b and CD10. Abbas p.15 Neutrophils live only 1–2 days in tissues. Abbas p.15 They respond fast because they activate preformed enzymes. Macrophages respond more slowly, and often need new gene transcription. Abbas p.15 Neutrophils can also extrude their nuclear contents as neutrophil extracellular traps (NETs). NETs immobilize and kill extracellular microbes, but they can also damage healthy tissue. Abbas p.15

> **Clinical correlation**
>
> Defects of neutrophil killing and migration cause recurrent bacterial and fungal infections. Chronic granulomatous disease (CGD) is caused by defective phagocyte oxidase and reactive oxygen species. Abbas p.487 Leukocyte adhesion deficiency type 1 (LAD-1) is caused by mutations in CD18, the β chain of β2 integrins. Abbas p.487 Module M4 covers these defects in detail.

3. Monocytes and macrophages

Two origins

Classical (inflammatory) monocytes make up 90–95% of human blood monocytes. They are rapidly recruited to sites of infection or injury, where they become macrophages. Abbas p.15-16 Most tissue-resident macrophages have a different origin. They come from yolk sac or fetal liver precursors, live for a long time and renew themselves. Abbas p.15 Examples are Kupffer cells (liver), alveolar macrophages (lung) and microglia (brain). Abbas p.15 Human macrophage markers include CD11b, CD14, CD68, CD16, CD64 and CCR5. Abbas p.15

Functions

- Cytokine production: activated macrophages make TNF, IL-1, IL-6 and IL-12. Abbas p.19 (Phagocytosis and killing: see M4.)
- Removal of apoptotic cells before they release their contents (efferocytosis). Abbas p.16
- Protection of healthy cells: the inhibitory receptor SIRPα binds CD47, a "don't eat me" signal on healthy cells. Abbas p.19

Classical (M1) versus alternative (M2) activation

Microbial products such as lipopolysaccharide and NK cell-derived interferon-γ (IFN-γ) activate macrophages. Abbas p.19 Classical activation gives microbicidal M1 macrophages. Alternative activation gives M2 macrophages, which promote tissue remodeling and repair. Abbas p.19 In Abbas Fig. 2.4, cytokines such as IL-4 and IL-13 drive collagen and angiogenic-factor production, which leads to repair and fibrosis. Abbas p.19 Many macrophage subsets do not fit neatly into M1 or M2. Abbas p.19

	M1 (classical)	M2 (alternative)
Typical stimuli	IFN-γ, microbial products (e.g., lipopolysaccharide) Abbas p.19	IL-4, IL-13 (Abbas Fig. 2.4) Abbas p.19
Main job	Kill microbes; inflammation Abbas p.19	Tissue remodeling, repair, fibrosis Abbas p.19

> **Analogy**
>
> An M1 macrophage is a soldier that destroys invaders. An M2 macrophage is a builder that repairs the damage after the fight.

Not covered in provided sources — needs expert input: specific surface markers used to tell M1 from M2 macrophages in humans (Abbas refers to Chapter 10, which is not in the source set).

4. Dendritic cells: the bridge to adaptive immunity

DCs are sentinels in epithelia and tissues. They capture antigens and carry them to lymph nodes, where naive T cells circulate. In this way they link innate and adaptive immunity. Abbas p.20 DCs activated by pathogen-associated molecular patterns (PAMPs) express costimulators. They also direct naive T cells to become effector subsets such as Th1 or Th17 cells. Abbas p.81

Subset	Key features	Main role
cDC2	Most numerous DC Abbas p.21	Presents antigen mainly to CD4+ T cells Abbas p.21
cDC1	IRF8, BATF3; CD141, CLEC9A, XCR1; makes IL-12 Abbas p.21-23	Cross-presents antigen to naive CD8+ T cells; induces Th1 responses Abbas p.21-23
Plasmacytoid DC (pDC)	BDCA-2 (CD303), CD123; high TLR7 and TLR9; transcription factor E2-2 Abbas p.23	Main source of antiviral type I IFN Abbas p.22
Langerhans cell	Epidermis; Birbeck granules; langerin (CD207); embryonic origin like tissue macrophages Abbas p.22	Antigen presentation to CD4+ T cells or tolerance Abbas p.22

pDCs can make up to 1000 times more type I IFN than other cell types. Janeway p.111

> **Exam pearl**
>
> Follicular dendritic cells are NOT dendritic cells in the immunological sense. They are stromal cells, they do not come from bone marrow, and they do not present protein antigens to T cells. Abbas p.22

5. Natural killer (NK) cells

Identity

NK cells are 5–20% of lymphocytes in blood and spleen. They are abundant in liver and placenta. Abbas p.82 In blood they are CD56-positive and CD3-negative, and most express CD16. Abbas p.82 NK cells kill without prior activation and use germline-encoded receptors, not clonal antigen receptors. Abbas p.82

Two effector functions

1. The NK cell kills an infected or stressed cell. Perforin helps granzymes enter the target cell. Granzymes then trigger apoptosis. Abbas p.82
2. The NK cell secretes IFN-γ. IFN-γ activates macrophages to destroy the microbes that they have phagocytosed. Abbas p.82

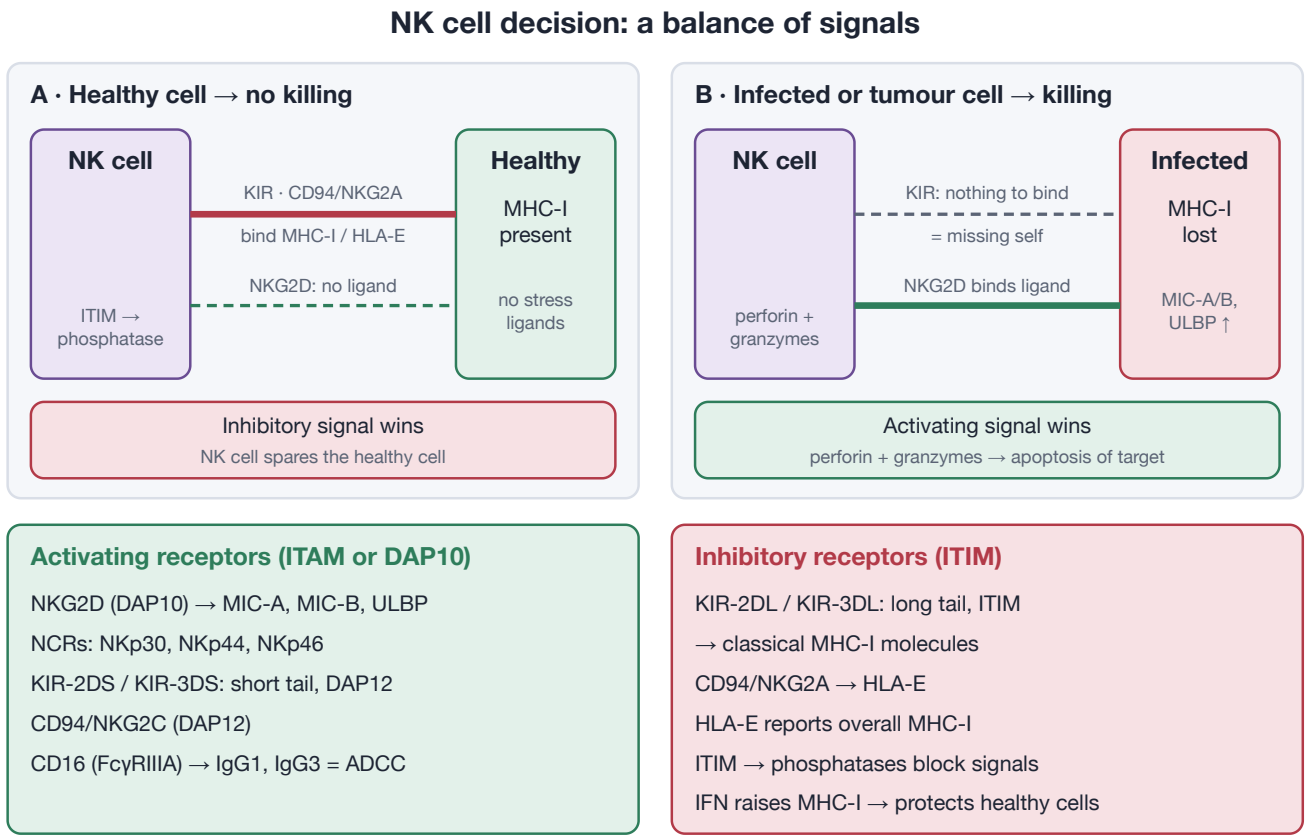
NK cells act early. They kill virus-infected cells before antigen-specific cytotoxic T lymphocytes (CTLs) become fully active after 5–7 days. Abbas p.82 IL-12, IL-15, IL-18 and type I IFNs increase NK killing and IFN-γ secretion. IL-15 is also an NK growth factor. Abbas p.85

The balance: activating versus inhibitory receptors

Each NK cell adds up signals from activating and inhibitory receptors. The balance decides the outcome. Abbas p.83

- **Inhibitory receptors** recognise MHC class I (MHC-I). Inhibitory KIRs bind classical MHC-I molecules. The CD94/NKG2A heterodimer binds HLA-E, which displays peptides from other MHC-I molecules. Abbas p.85 Inhibitory receptors carry ITIMs, which recruit phosphatases that block activating signals. Abbas p.85
- **Activating receptors** signal through ITAMs. Abbas p.85 NKG2D binds the stress proteins MIC-A, MIC-B and ULBP on infected and tumour cells, and signals through DAP10. Abbas p.83 The natural cytotoxicity receptors (NCRs) NKp30, NKp44 and NKp46 are also activating. Janeway p.117
- **Same family, opposite signals.** Inhibitory KIRs (KIR-2DL, KIR-3DL) have long tails with ITIMs. Activating KIRs (KIR-2DS, KIR-3DS) have short tails and pair with DAP12. Janeway p.115-116 NKG2A is inhibitory, but NKG2C pairs with DAP12 and is activating. Janeway p.116

Missing self. Viruses, cell stress and many cancers reduce MHC-I on the cell surface. The NK cell then loses its inhibitory signal and is activated. This is called recognition of missing self. Abbas p.85 Interferons increase MHC-I on uninfected cells and so protect them from NK cells. Janeway p.115



D6. NK cell "missing self". The NK cell integrates activating and inhibitory signals. Loss of MHC-I removes inhibition; stress ligands add activation. CD16 lets NK cells kill IgG-coated cells (ADCC). Fact IDs M2-F049, M2-F054, M2-F055, M2-F050, M2-F051, M2-F056, M2-F057, M2-F052, M2-F053, M2-F058, M2-F046, M2-F048. Abbas p.83 Abbas p.85 Janeway p.115-116 Janeway p.117

CD16 and ADCC

CD16 (FcγRIIIA) is a low-affinity receptor for IgG. It binds IgG1 and IgG3 on antibody-coated cells. The NK cell then kills the coated cell. This is antibody-dependent cell-mediated cytotoxicity (ADCC). Abbas p.84

> **Clinical correlation: NK cell deficiency**
>
> Patients with numerical or functional NK cell deficiency have severe herpesvirus and papillomavirus infections. Abbas p.488 Autosomal dominant GATA2 mutations cause loss of NK cells, together with low monocytes, DCs and B cells. Abbas p.488 Autosomal recessive CD16 mutations cause a functional NK cell deficiency. Abbas p.488 In Chédiak-Higashi syndrome, NK function is poor because the cytotoxic granules are abnormal. Abbas p.488

6. Mast cells, basophils and eosinophils

These three myeloid cells all have granules full of inflammatory mediators. All three take part in defence against helminths and in allergic disease. Abbas p.20 They differ in location, life span, receptors and mediators. Abbas p.466

Feature	Mast cell	Basophil	Eosinophil
Where	Tissues; near vessels, nerves, under epithelia; not in blood Abbas p.465	Blood (0.5% or less of leukocytes); recruited to tissues Abbas p.465	Blood; normally also in respiratory, gut and genitourinary mucosa Abbas p.20
Life span Abbas p.466	Weeks to months	Days	Days to weeks
Main growth factor Abbas p.466	SCF, IL-3	IL-3	IL-5
FcεRI Abbas p.466	High (αβγ2)	High (αβγ2)	Low (αγ2), no proven IgE activation Abbas p.467; Abbas p.473
Main granule contents	Histamine; heparin and/or chondroitin sulfate; neutral proteases (tryptase and/or chymase) Abbas p.466; Abbas p.467	Histamine; chondroitin sulfate; protease Abbas p.466	MBP, ECP, eosinophil peroxidase Abbas p.473; galectin-10 (preformed in cytosol) Abbas p.467
New mediators	PGD2, leukotrienes C4/D4/E4, PAF; cytokines Abbas p.467		PAF, prostaglandins, cysteinyl leukotrienes; cytokines (e.g., IL-4, TGF-β) Abbas p.473

Mast cells

Stem cell factor (SCF) binds c-KIT on mast cell precursors and is essential for mast cell development. Abbas p.465 FcεRI binds IgE with very high affinity. As a result, most mast cells carry IgE even in nonatopic people. Abbas p.466 On mast cells and basophils, FcεRI is an αβγ2 tetramer. Abbas p.467

IgE-mediated activation follows this sequence:

1. Multivalent allergen binds IgE that sits on FcεRI. Abbas p.468
2. The allergen cross-links the FcεRI molecules. Abbas p.468
3. The mast cell releases preformed granule contents (degranulation). Abbas p.468
4. The mast cell makes new lipid mediators and cytokines. Abbas p.468

Mast cells can also be activated without IgE. Complement fragments C3a and C5a (the anaphylatoxins) cause degranulation. TLR ligands, ATP and cytokines such as SCF and IL-33 also activate mast cells. They cause the production of lipid mediators and cytokines, but not degranulation. Abbas p.470 Note: Abbas Chapter 4 states that TLR ligands can induce degranulation. Abbas p.86 The receptor MRGPRX2 on skin mast cells binds some antibiotics, anaesthetic drugs and venom components. It is a likely mediator of some allergy-like drug reactions. Abbas p.470

> **Clinical correlation: tryptase and mastocytosis**
>
> Tryptase is present in all human mast cells and in no other known cell type. For this reason, serum tryptase is a marker of mast cell activation and is used to diagnose anaphylaxis. Abbas p.471 Human mast cell subsets contain tryptase only, or tryptase and chymase. Abbas p.465 Systemic mastocytosis (SM) is a clonal mast cell disorder associated with c-KIT mutations and increased mediator release, including tryptase. Abbas p.470-471

Basophils

Basophils are blood granulocytes. They are normally absent from tissues but can be recruited to inflammatory sites. Abbas p.465 Like mast cells, basophils express FcεRI, bind IgE and are triggered when antigen binds the IgE. Abbas p.465

Eosinophils

GM-CSF, IL-3 and IL-5 promote eosinophil maturation from myeloid precursors. Abbas p.20 Eosinophils carry Fc receptors for IgA and IgG, TLRs and IL-5 receptors. Abbas p.20 IL-5 from Th2 cells and ILC2s increases eosinophil production and activates mature eosinophils. Abbas p.473 Eotaxin (CCL11) binds CCR3 on eosinophils and recruits them into tissue. Abbas p.473

Major basic protein (MBP) and eosinophil cationic protein (ECP) damage helminths, bacteria and host cells. Abbas p.473 Eosinophil peroxidase is different from neutrophil MPO. It makes hypochlorous or hypobromous acid. Abbas p.473 Eosinophils make large amounts of galectin-10. Galectin-10 forms Charcot-Leyden crystals at sites of chronic type 2 inflammation, such as asthmatic bronchi. Abbas p.473

In the late-phase reaction, neutrophils, Th2 cells and eosinophils enter the tissue a few hours to 2 days after allergen exposure. Abbas p.465

> **Exam pearl**
>
> Eosinophils have very little FcεRI, and their receptor lacks the signalling β chain. There is no convincing evidence that allergen bound to IgE activates eosinophils. Abbas p.467; Abbas p.473 The cells that respond directly to IgE cross-linking are mast cells and basophils. Abbas p.466; Abbas p.465

Not covered in provided sources — needs expert input: basophil activation markers used in the basophil activation test, and eosinophil surface markers other than CCR3 and IL-5 receptor.

7. Innate lymphoid cells (ILCs)

ILCs are bone marrow-derived lymphoid cells without T-cell receptors (TCRs). Abbas p.81 They secrete cytokines like those of helper T cells. They live mainly in mucosal tissues such as lung and intestine, and are rare in blood. Abbas p.30 A common ILC precursor is identified by the transcription factor Id2. Abbas p.81

Subset	Transcription factor Abbas p.81	Activated by Abbas p.81	Main cytokines Abbas p.81	Function Abbas p.81-82
ILC1	T-BET	IL-12, IL-15, IL-18	IFN-γ	Defence against intracellular microbes
ILC2	GATA3	IL-25, IL-33, TSLP	IL-5, IL-13	Defence against helminths; allergic disease
ILC3	RORγt	IL-1, IL-23	IL-22 and/or IL-17	Defence against extracellular fungi and bacteria; epithelial barrier

NK cells resemble ILC1s because both secrete IFN-γ. However, ILC1s are not cytotoxic. Abbas p.82

ILC2s in allergy

1. Epithelial barrier cells release IL-25, IL-33 and TSLP. IL-33 is stored in epithelial nuclei. Damaged epithelial cells release it rapidly, so it is called an alarmin. Abbas p.94
2. These cytokines stimulate ILC2s, Th2 cells and mast cells to make IL-4, IL-5 and IL-13. Abbas p.94
3. ILC2s already sit in the tissue. Their cytokines can start allergic inflammation before Th2 cells arrive. Abbas p.465
4. Later, ILC2s work together with Th2 cells to sustain the inflammation. Abbas p.465

> **Analogy**
>
> ILC2s are the local fire brigade that is already in the building. Th2 cells are the reinforcements that arrive days later with more equipment.

8. Innate-like lymphocytes

Some T and B cells have rearranged antigen receptors of limited diversity. These include invariant natural killer T (iNKT) cells, γδ T cells, mucosa-associated invariant T (MAIT) cells and B-1 cells. They behave more like innate cells than adaptive cells. Abbas p.85-86

- **MAIT cells** recognise bacterial riboflavin metabolites. They make up almost 50% of liver lymphocytes. Abbas p.24
- **iNKT cells** recognise glycolipids presented by CD1. They rapidly secrete IL-4, IL-10 and IFN-γ. Janeway p.119
- **Intraepithelial γδ T cells** recognise target molecules directly, not as peptide on MHC. Janeway p.119
- **B-1 cells** make IgM against polysaccharides within 48 hours, without T-cell help. In mice, they protect against *Streptococcus pneumoniae*. Janeway p.119

9. Cell ID cards

Innate cell ID cards

Neutrophil

CD11b, CD18, CD15, CD66b
Azurophilic: MPO, defensins
Specific: lysozyme, elastase
Fast phagocyte; NETs; 1–2 d
Disease: CGD, LAD-1 (→ M4)

Macrophage

CD14, CD68, CD64, CD16, CCR5
Makes TNF, IL-1, IL-6, IL-12
M1: microbicidal (IFN-γ, LPS)
M2: repair, fibrosis (IL-4/13)
SIRPα–CD47: “don’t eat me”

Dendritic cell

All: CD11c; need FLT3L
cDC1: CD141, XCR1 → CD8 T
cDC2: most numerous → CD4 T
pDC: CD123, TLR7/9 → type I IFN
Disease: ↓ in GATA2 deficiency

NK cell

CD56+ CD3–; most CD16+
Perforin + granzymes
KIR, NKG2A, NKG2D, NCRs
Kills infected/tumour cells
Disease: GATA2, CD16, CHS

Mast cell

FcεRI high; c-KIT; MRGPRX2
Histamine, heparin, tryptase
Makes PGD2, LTC4, PAF
Tissue-resident; weeks–months
Disease: anaphylaxis, SM

Basophil

FcεRI high; IL-3
Histamine, chondroitin sulfate
Blood: <1% of leukocytes
Life span: days
Recruited to inflamed sites

Eosinophil

IL-5R, CCR3; FcεRI very low
MBP, ECP, eosinophil peroxidase
Galectin-10 → Charcot-Leyden
Anti-helminth; late phase
Disease: allergic asthma

ILC1 · ILC2 · ILC3

No TCR; mucosal tissues
ILC1: T-BET → IFN-γ
ILC2: GATA3 → IL-5, IL-13
ILC3: RORγt → IL-17, IL-22
ILC2 ← IL-25, IL-33, TSLP

Innate-like lymphocytes

γδ T, iNKT, MAIT, B-1 cells
Low-diversity TCR or BCR
MAIT: riboflavin metabolites
iNKT: glycolipid on CD1
B-1: fast IgM, no T-cell help

D5. Innate cell "ID cards". Each card lists markers or receptors, granules or products, main job and a key disease. SM = systemic mastocytosis; CHS = Chédiak-Higashi syndrome. Fact IDs M2-F017, M2-F016, M2-F015, M2-F020, M2-F018, M2-F105, M2-F106, M2-F025, M2-F032, M2-F028, M2-F029, M2-F030, M2-F027, M2-F009, M2-F035, M2-F034, M2-F036, M2-F060, M2-F043, M2-F046, M2-F054, M2-F050, M2-F051, M2-F045, M2-F061, M2-F062, M2-F067, M2-F064, M2-F075, M2-F072, M2-F077, M2-F073, M2-F065, M2-F066, M2-F079, M2-F012, M2-F083, M2-F088, M2-F090, M2-F085, M2-F086, M2-F087, M2-F091, M2-F092, M2-F094, M2-F095, M2-F100, M2-F101, M2-F102, M2-F104.

Abbas p.15 Abbas p.21-23 Abbas p.82 Abbas p.467 Abbas p.473 Abbas p.81 Abbas p.85-86 Janeway p.119

5 key points

1. All innate cells come from the HSC. Granulocytes, monocytes/macrophages, DCs and mast cells are myeloid; NK cells and ILCs are lymphoid. Abbas p.30; Abbas p.32
2. Most tissue-resident macrophages come from fetal precursors. Classical (M1) activation kills microbes; alternative (M2) activation repairs tissue. Abbas p.15; Abbas p.19
3. pDCs are the main source of type I IFN; cDC1s cross-present to CD8+ T cells; cDC2s present to CD4+ T cells. Abbas p.22; Abbas p.21-23; Abbas p.21
4. NK cells kill when activating signals win. Loss of MHC-I ("missing self") removes the inhibition by KIRs and CD94/NKG2A. Abbas p.83; Abbas p.85
5. Mast cells and basophils have high FcεRI; eosinophils have low FcεRI. ILC2s (GATA3; IL-5, IL-13) respond to IL-25, IL-33 and TSLP and start allergic inflammation early.

Abbas p.466; Abbas p.81; Abbas p.94; Abbas p.465

3 exam traps

1. Tryptase, not histamine, is the mast cell activation marker: tryptase is in all human mast cells and no other known cell. Chymase is only in some mast cells. Abbas p.471; Abbas p.465
2. CD16 on NK cells binds IgG (IgG1, IgG3), not IgE. The IgE receptors that trigger allergy are FcεRI on mast cells and basophils. Abbas p.84; Abbas p.466
3. Follicular dendritic cells are stromal cells and do not present protein antigens to T cells. Do not confuse them with conventional DCs. Abbas p.22

Practice: M2 questions in the quiz (IDs M2-A1 … M2-K2).

Sources: Abbas, Cellular and Molecular Immunology, 11th ed., pp. 14–24, 30–32, 81–86, 94, 101, 465–473, 487–488; Janeway's Immunobiology, 8th ed., pp. 76–77, 111, 113–119. · Fact IDs refer to `ledger/facts.csv`.